

# Towards Autonomous Air Cargo Network Design: A Measured Matheuristic over Branch-and-Price at Integrator Scale

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Code, instance generators and experiment logs:  
[https://github.com/csotogd/flight\\_scheduling](https://github.com/csotogd/flight_scheduling)

## Abstract

Integrated airline schedule design — which flights to operate, which fleet flies each, how aircraft rotate, how cargo routes — faces two obstacles to automation at the scale of a global express integrator. The candidate flight space cannot be handed over by a planner nor enumerated exhaustively once the network spans a hundred-plus stations over a periodic week; and the integer layer of the exact method stalls entirely there. We present an approach that addresses them together, and a working system that measures it. Building on the branch-and-price-and-cut of Derigs and Friederichs (OR Spectrum 35:325–362, 2013), an autonomous design loop generates its own candidate flights round by round — propose, re-optimize, keep what pays, evict what does not — while a matheuristic layer keeps the optimization moving at scale: a *deliver-all* service model that prices contracted delivery as recourse and makes every subproblem feasible by construction; greedy cover seeding monetized by one warm LP over freshly priced columns; an escalating local-branching ball for integer adoption; and a geographic fix-and-optimize decomposition whose cross-boundary shipments carry *exact* connection windows read from the frozen timetable, so regional improvements splice into a globally verified schedule and the cycle is monotone by construction. On a synthetic integrator-scale instance (29,819 shipments, 1,250 flights, 154 aircraft), the whole-network integer search adopts nothing in ninety minutes while the regional cycle gains 4–5 M of weekly profit (synthetic units) in minutes; every component is toggleable and measured by ablation, including three negative results reported with equal prominence. Results are single-seed and exploratory; we state this, and the road to a full battery, explicitly.

## 1 Introduction

An express integrator sells a promise — everything tendered tonight is delivered across the planet within days — and keeps it with one of the most tightly coupled logistics systems in existence: hub-and-spoke networks whose feeders, overnight hub sorts, intercontinental trunks and morning distribution waves must interlock to the minute, flown by a heterogeneous fleet whose every aircraft chains through the week in a closed rotation. Planning such a network is traditionally cut into layers — which flights to operate, which fleet flies each, how aircraft rotate, how shipments route — optimized separately and reconciled by hand. The layers, however, interact strongly: a flight is worth operating only if cargo can reach it, cargo can reach it only if rotations position an aircraft, and an aircraft is free only if the rest of the

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schedule releases it. Sequential planning leaves structural money on the table, which is the classic argument for integrated optimization [15, 16].

Derigs and Friederichs [1] made that argument concrete for air cargo: they formulated the integrated problem (ACSP) over a cyclic week, gave a path/string column formulation, and solved it exactly by branch-and-price-and-cut (B&P&C) with problem-specific branching and implied-bound cuts, on four synthetic airline archetypes with up to a few thousand shipments.

Two obstacles stand between that method and unattended use at integrator scale. The first is the candidate space: under the “pragmatic paradigm” the model *chooses from* a flight list an expert planner must write. Recent work avoids the planner by enumerating every flight slot of a discretized time-space network [17] — tractable on a 24-airport, single-night network, but not over a hundred-plus stations and a periodic week, where the slot space explodes. Either way, the creative half of network design does not scale by itself. The second obstacle is empirical. It is not a gradual degradation: on the set-I instance of each of our families the base method closes gaps under 2% up to nine thousand shipments, and at GI scale needs a single branch-and-bound node to do so (Section 8.1); larger sets of the same families can sit just above that line (MI at 4,000 shipments on its set-II instance reaches 2.2% in ten minutes). One family further up, the picture changes qualitatively, and we establish it twice, independently, in Section 8.2: at integrator scale — tens of thousands of distinct origin–destination shipments per week over a thousand-plus flights, an order of magnitude above the literature’s instances — the *integer* layer of the exact method stalls completely. From a good incumbent, ninety further minutes of whole-network branch-and-price explore one to three nodes and adopt nothing; a planner waiting for the tool gets silence. Figure 1 shows the anatomy behind the scale: most of an integrator’s shipments are tiny and scattered, which drives much of what follows.

This paper brings an approach that attacks both bottlenecks together, so that network design can run *autonomously*: a design loop that generates, tests and retires its own candidate flights — no planner-written list — wrapped around a matheuristic layer that keeps the underlying optimization adopting improvements at a scale where the exact method alone has gone quiet. The planner’s role moves from feeding the tool to reviewing its output.

Our answers are empirical. We reimplemented the full method of [1] (Section 2), generated instances from reimplementations of the paper’s four archetype recipes plus a new public integrator-scale family, and then measured our way through a sequence of additions, keeping what worked and reporting what did not. The result is a matheuristic in the modern sense [13, 14]: heuristic scaffolding — greedy construction, large-neighborhood integer moves, spatial fix-and-optimize — around an exact column-generation engine that supplies the three things heuristics cannot: feasible *columns* for a timed, capacitated network; *dual prices* that tell every heuristic where value lies; and *valid bounds* that certify how much is left on the table.

**Related work: the application.** Air cargo operations have a rich literature — see the reviews of Feng et al. [16] and Brandt and Nickel [23] — and the planning layers have traditionally been treated separately, in the tradition of service network design [15]. Integrated formulations have a lineage of their own: Derigs et al. [18] combine flight selection with rotation planning; Derigs and Friederichs [1] add fleet assignment and solve the cyclic-week problem exactly by B&P&C; Xiao et al. [19] integrate tail assignment and cargo routing with through-cargo connections; Yildiz and Savelsbergh [20] optimize package express operations with single-flight shipments; Zheng et al. [21] add hub stay times with an ALNS metaheuristic; and Huang et al. [22] recover disrupted cargo schedules with machine-learning-guided column-and-row generation.

The closest work to ours is concurrent. Zhu et al. [17] jointly determine cargo flight schedules, fleet routes and cargo allocation — including leased belly capacity and through-cargo connections — and solve the result with a column-generation heuristic that finishes on a restricted MIP over an expanded route pool, validated on data from an industry partner. Our work shares

that overall shape (an integrated route-based model, column generation, a restricted-master finish) and we claim no priority over it. Three differences frame what we add. (i) *Horizon*: their model spans a single overnight window with aircraft routes of at most three flights; ours is a periodic week in which rotations chain across the entire horizon, which is precisely what forces the balance-repair and fleet-slicing machinery of Section 6. (ii) *Candidate space*: they too need no planner-written list — they enumerate every flight slot of a discretized time-space network — but that enumeration is tractable because their networks span 24 airports and 257 commodities; over 150 airports and 29,819 weekly shipments it is not, so the candidate space must instead be *sampled adaptively*, which is what the design loop of Section 5 does. (iii) *Scale and evidence*: their large-scale instances carry 257 commodities and a 41–54 aircraft fleet, two orders of magnitude below the shipment counts we target; conversely they report real operational data and CPLEX and Benders baselines, all three of which we lack and list among our limitations (Section 9).

**Related work: the instruments.** Methodologically we compose established components, each adapted to this problem’s structure. Column generation and branch-and-price [2, 3, 4, 24] supply priced itineraries, dual signals and valid bounds; under truncation we bound with Farley’s construction [9], and against early dual oscillation we implement Wentges smoothing [7] in the stabilized-colgen tradition of du Merle et al. [8] — made mispricing-safe so certificates never degrade. For integer progress on masters too large to re-solve, we use local branching [5], a cousin of relaxation-induced neighborhoods [6], escalating the ball radius when flat. Our geographic decomposition belongs to the large-neighborhood and fix-and-optimize families [10, 11, 12], transplanted from variable space to *geography*; its novelty is what the cyclic weekly schedule forces: exact gateway windows read from the frozen timetable, balance-repaired partitions so both sides still assemble into rotations, and a monotone, independently verified merge. Our contribution is accordingly not a new component but a measured *composition*: which instruments still work when the master holds thirty thousand commodities, how they must be adapted, and — reported with equal prominence — which plausible instruments do *not* pay.

## Contributions.

1. An *autonomous design loop* (Section 5) for a candidate space too large to enumerate: four demand-driven generator classes, per-round solve–score–evict with amnesty, and the scale devices (demand consolidation, zone rotation) that keep rounds tractable — so the active model stays bounded while the explored candidate space grows without limit, the integrated model designs the network unattended, and the planner reviews the result.
2. A *deliver-all* service model for integrators (Section 3): demand rows become equalities and every shipment carries an always-available contracted-delivery column priced at a multiple of own flying economics, derived from the mandatory schedule only so the tariff is identical across all model variants. Every relaxation, sub-MIP and decomposition block becomes feasible by construction, and the optimization becomes “own network versus contracting bill”.
3. An integer-adoption pipeline for huge masters (Section 4): greedy cover seeding, flow monetization by a single warm LP over the freshly generated column pool, and an escalating local-branching ball [5] as the primal heuristic, with a per-round observable (*LP promise vs. integer adoption*) that separates “the candidate generator is weak” from “integer adoption is the bottleneck”.
4. A geographic fix-and-optimize decomposition (Section 6) whose distinguishing features are *exact* gateway windows read from the frozen timetable (no estimated connection times), a balance-repair partition that keeps both the frozen and the re-optimized side

assemblable into rotations, an exact fleet slice with a learned correction for chain-splitting rounding, and a monotone merge guard: a block result is accepted only when the independently verified *global* schedule improves. An optional pairwise “relay” re-opens cross-region demand with no cross-pass bookkeeping.

5. A measured engineering layer (Section 7): a bit-identical parallel pricing sweep, Farley-type bounds [9] under truncation with a membership filter, and per-iteration cost breakdowns that make the next bottleneck visible instead of guessed. Mispricing-safe dual smoothing [7, 8] is implemented and correctness-verified; its performance evaluation belongs to the planned battery and no benefit is claimed here.
6. An honest computational study (Section 8) on public, deterministic, regenerable instances, including three negative results: coordinated “hub wave” candidate bundles that the optimizer never adopts; cross-region relay passes that run clean but earn nothing at the chosen recourse tariff; and a specialized LP method (sifting) that loses to warm-started dual simplex on the colgen master.

We are explicit about scope (Section 9): headline numbers are single-seed, the instances are synthetic with AI-estimated parameters, and comparability with [1] is qualitative because neither their instances nor their code were published. This preprint plants the system, the mechanisms and the measurement discipline; the multi-seed battery and an arc-flow baseline are the declared next step. The remainder of the paper follows the pipeline: model and base algorithm (Sections 2–3), integer adoption and network design (Sections 4–5), geographic decomposition (Section 6), engineering (Section 7), results and limitations (Sections 8–9).

## 2 Problem and base algorithm

We summarize [1] only to fix vocabulary. A weekly cyclic horizon of  $N = 10,080$  minutes carries: *airports* (some transfer hubs with minimum connection times, transfer and storage costs); *fleets* with counts, payload/volume capacities and costs; *flights* (sequences of legs; mandatory or optional; possibly external capacity with booking costs); and *shipments* (O&D demands with release times, delivery deadlines, tonnage, volume, freight rate).

Model ACSP-T is a path/string formulation over a time-space network: columns are cargo *paths* (feasible timed itineraries for one O&D) and flight *strings* (fleet-assigned flight sequences; in the FARP-T variant used throughout this paper a string is a single flight and rotations are chained through ground arcs of the timeline). Columns are priced by two oracles: PRICE-P, a resource-constrained shortest path solved with A\*; and PRICE-S for strings. Implied-bound cuts link flow to selection; problem-specific branching (flight in/out, fleet-flight, follow-on) keeps both pricers intact. Without maintenance the procedure is exact. Our reimplementations follow the paper; every accepted incumbent is re-validated by an independent feasibility checker that shares no code with the solver.

### 2.1 The master problem, written down

Table 1 fixes notation; we state the restricted master in the form our solver actually builds, since every mechanism in Sections 3–6 is defined as an operation on these rows and columns.

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|                            |                                                                               |
|----------------------------|-------------------------------------------------------------------------------|
| <i>Sets</i>                |                                                                               |
| $A, H \subseteq A$         | airports; transfer hubs                                                       |
| $K$                        | fleet types, $n_k$ aircraft of type $k$ , volume capacity $V_k$               |
| $F$                        | flights: $F^M$ mandatory, $F^O$ optional, $F^X$ external                      |
| $L, L_f$                   | legs; legs of flight $f$                                                      |
| $Q$                        | shipments (O&D demands), $d_q$ tonnes, rate $r_q$ , $\nu_q$ m <sup>3</sup> /t |
| $P_q, S$                   | feasible cargo paths of $q$ ; feasible flight strings                         |
| $\mathcal{E}, \mathcal{G}$ | timeline events; ground arcs                                                  |
| <i>Parameters</i>          |                                                                               |
| $W_k(l)$                   | payload of fleet $k$ on leg $l$ : the payload-range frontier (Section 2)      |
| $m_p$                      | margin of path $p$ : $r_q$ minus variable, transfer and storage cost          |
| $c_s, \chi_s$              | cost of string $s$ ; aircraft-periods it consumes                             |
| $\kappa_k, \varphi_f$      | fixed cost per aircraft of type $k$ ; booking cost of external flight $f$     |
| $\rho_q$                   | contracted-delivery cost per tonne of $q$ (Section 3)                         |
| <i>Variables</i>           |                                                                               |
| $y_p \geq 0$               | tonnes of $q$ routed on path $p \in P_q$                                      |
| $x_s \in \{0, 1\}$         | string $s$ selected                                                           |
| $g_a \geq 0$               | flow on ground arc $a$                                                        |
| $b_f \in \{0, 1\}$         | external flight $f$ booked                                                    |
| $z_q \geq 0$               | tonnes of $q$ delivered by contracted carrier                                 |

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Table 1: Notation.

$$\begin{aligned} \max \quad & \sum_{q \in Q} \sum_{p \in P_q} m_p y_p + \sum_{q \in Q} (r_q - \rho_q) z_q - \sum_{s \in S} (c_s + \chi_s \kappa_{k(s)}) x_s \\ & - \sum_{a \in \mathcal{G}} \chi_a \kappa_{k(a)} g_a - \sum_{f \in F^X} \varphi_f b_f \end{aligned} \quad (1)$$

$$\text{s.t.} \quad \sum_{p \in P_q} y_p + z_q = d_q, \quad \forall q \in Q \quad (2)$$

$$\sum_{p: l \in p} y_p - \sum_{s: l \in s} W_{k(s)}(l) x_s - \bar{w}_l b_{f(l)} \leq 0, \quad \forall l \in L \quad (3)$$

$$\sum_{p: l \in p} \nu_{q(p)} y_p - \sum_{s: l \in s} V_{k(s)} x_s - \bar{v}_l b_{f(l)} \leq 0, \quad \forall l \in L \quad (4)$$

$$\sum_{s: f \in s} x_s = 1 \quad \forall f \in F^M, \quad \sum_{s: f \in s} x_s \leq 1 \quad \forall f \in F^O \quad (5)$$

$$\sum_{s \in \delta^+(e)} x_s - \sum_{s \in \delta^-(e)} x_s + \sum_{a \in \delta^+(e)} g_a - \sum_{a \in \delta^-(e)} g_a = 0, \quad \forall e \in \mathcal{E} \quad (6)$$

$$\sum_{s: k(s)=k} \chi_s x_s + \sum_{a: k(a)=k} \chi_a g_a \leq n_k, \quad \forall k \in K \quad (7)$$

$$\sum_{p \in P_q: f \in p} y_p - d_q \sum_{s: f \in s} x_s \leq 0, \quad \forall (q, f) \in \mathcal{C} \subseteq Q \times F^O \quad (8)$$

Objective (1) is profit: path margins and contracted deliveries against string, aircraft, and external booking costs. Rows (2) are the demand rows — equalities under deliver-all (Section 3),  $\leq d_q$  otherwise, in which case  $z_q$  is absent. Rows (3)–(4) couple cargo flow to capacity: a leg carries tonnage only if a string covers its flight, and the capacity offered is the payload-range frontier  $W_k(l)$ , which is a constant per  $(l, k)$  and therefore keeps the coupling linear ( $\bar{w}_l, \bar{v}_l$  are the posted capacities of external legs). Rows (5) cover mandatory flights exactly once and optional flights at most once; (6) is per-fleet flow balance at timeline events, which is what

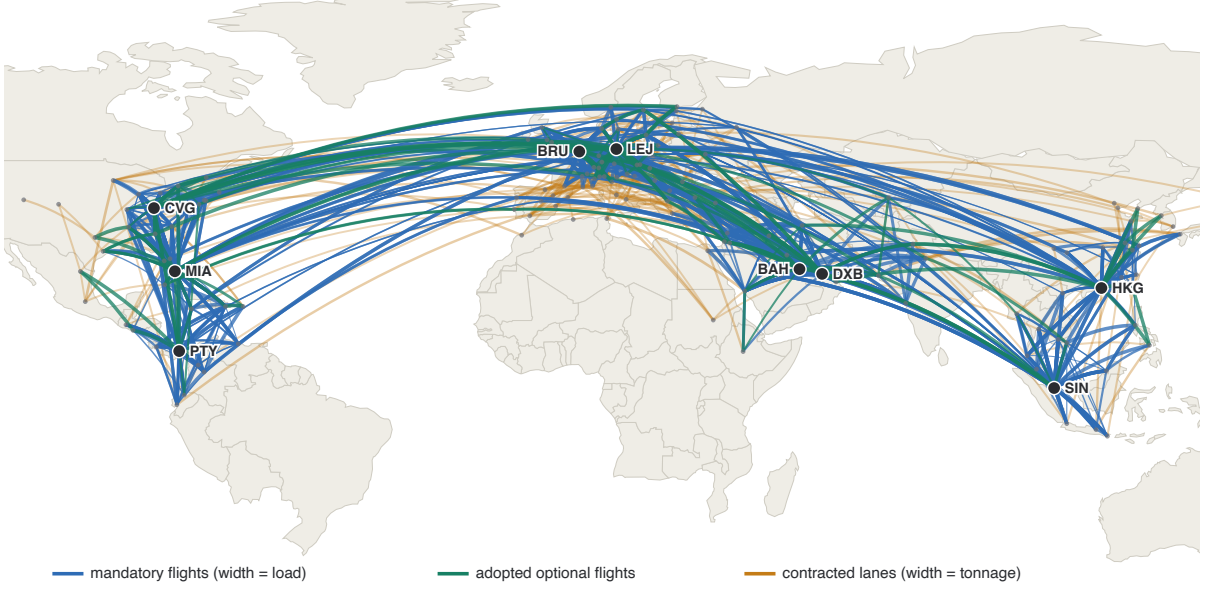


Figure 1: The best full-demand schedule on RLA-I-s1, drawn to scale. Blue: mandatory flights (width = load); green: adopted optional flights; amber, underneath: the 250 heaviest *contracted* lanes (width = tonnage) — demand served by paid external carriers. The amber fans around the hubs are the 46.6 M contracting bill of Section 8.3 made visible, and the target of the wave, relay and decomposition experiments.

|                                                 |                         |
|-------------------------------------------------|-------------------------|
| airports / transfer hubs                        | 150 / 9                 |
| fleets (aircraft)                               | 6 (154)                 |
| flights in base schedule (mandatory / optional) | 1,250 (625 / 625)       |
| shipments (O&D demands)                         | 29,819                  |
| total demand                                    | ~62.7 kt/week           |
| service commitment                              | deliver-all (Section 3) |

Table 2: The integrator-scale instance RLA-I-s1 (synthetic, seed 1).

makes selected strings chain into closed weekly rotations, and (7) caps aircraft. Rows (8) are the implied-bound cuts of [1], separated on the fly for the violated pairs  $\mathcal{C}$  of shipments and *own optional* flights. The restriction to  $F^O$  matters: for a mandatory flight the row is implied by (5) together with the demand row (2) (the flight is always covered, so the cut relaxes to  $\sum_p y_p \leq d_q$ ), and for an external flight — which no string covers — the right-hand side would vanish identically and the cut would wrongly force flow off legally booked external capacity ( $b_f = 1$ ), cutting feasible integer points.  $P_q$  and  $S$  are exponential and generated by pricing; the LP over the generated subset is the RMP throughout this paper.

**Instances.** We reimplemented the generation recipes of the four archetypes of [1] (regional, international, mixed, express) from the published parameter tables, and added two fictional families: GI (“global integrator”: 4 hubs, 140 aircraft, dense demand) and RLA (“real-life-scale airline”: 9 hubs on three continents, 154 aircraft in six fleets, gravity-model demand on 60% of ordered station pairs with weekly lane recurrence, 1,250-flight base schedule of which half is mandatory, and 29,819 shipments). All parameters are AI-estimated for plausibility; no real airline data is used. Generators are deterministic in an integer seed and public. Table 2 summarizes the instance RLA-I-s1 used for the scale experiments.

### 3 A deliver-all service model

Integrators do not cherry-pick profitable freight; a service commitment means everything tendered is delivered, if necessary by a contracted external carrier. We model this by (i) turning demand rows into equalities and (ii) giving every shipment *od* a *contracted delivery column* with objective coefficient  $r_{od} - c_{od}^{\text{rec}}$ , upper bound  $d_{od}$ , and a single nonzero entry in the demand row. The recourse tariff is

$$c_{od}^{\text{rec}} = \mu \cdot (\delta_{od} v + \delta_{od} \phi / W),$$

with  $\delta_{od}$  the great-circle distance,  $v$  the average own variable cost per tonne-km,  $\phi$  the median own fixed cost per km,  $W$  the largest fleet payload, and  $\mu = 3$  throughout. Crucially,  $v$  and  $\phi$  are derived from the *mandatory* flights only: that set is identical in every model variant of a design or decomposition run (optional candidates come and go), so the tariff — and hence the profit scale — is stable across all comparisons. An earlier version derived the tariff from all flights and suffered a  $\sim 1\%$  per-round drift that silently invalidated round-over-round comparisons.

Three consequences matter. Revenue becomes a constant, so the problem is a cost minimization (own flying vs. contracting bill). Every selection of flights is completable, so seeds, restricted sub-MIPs and decomposition blocks are feasible by construction — the entire pipeline of Sections 4 and 6 leans on this. And the recourse coefficient caps the demand duals at the tariff, a free mild stabilization of the master.

### 4 Integer adoption at scale

At integrator scale the practical failure is primal: branch-and-bound explores 1–3 nodes within realistic budgets (Section 8.2), so incumbents must come from elsewhere. Our pipeline has four stages, each measured in isolation in Table 4.

**(a) Greedy cover seed.** A constructive pass assigns every mandatory flight to its lightest compatible fleet as balanced round trips (inter-hub mirror pairs kept together), followed by an evaluated repair loop. The seed is feasible but flowless: it earns no revenue (deliver-all: everything is contracted).

**(b) Seed flow loader.** After the root column generation, the seed’s selection is fixed in the master and *one* warm LP optimizes cargo flows over the column pool the pricers just generated. This is where column generation pays its first dividend: the pool contains thousands of priced itineraries, and one LP monetizes the skeleton in seconds. No new pricing is needed.

**(c) Escalating local-branching ball.** The primal heuristic solves the restricted master MIP inside a Hamming ball of at most  $k$  selection flips around the incumbent [5], escalating  $k \rightarrow 2k \rightarrow 4k$  (default  $k = 60$ ) while flat within a budget slice. The incumbent is always feasible inside the ball, so the step is monotone; the solver, not a hand-built neighborhood, chooses *which* flips, which keeps cross-network rotation rewirings reachable. An unrestricted restricted-master MIP at this scale finds nothing in equal budget (Table 4); the ball turns the same budget into adopted profit. On adoption, flows are re-monetized by one further warm LP.

**(d) Promise vs. adoption.** Each design round logs the LP bound uplift contributed by the current candidate batch (*promise*) against the integer profit actually adopted. This single

observable separated two hypotheses we could not otherwise distinguish — “the proposer generates worthless candidates” vs. “the integer layer cannot digest them” — and later diagnosed a genuine economic no (Section 8.5).

## 5 The autonomous design loop

The base algorithm assumes a human writes the optional-flight candidate list; this section removes that assumption without enumerating the slot space. Each round, a proposer reads the previous solution’s gaps — unroutable demand and capacity-crowded shipments, ranked by revenue at risk — and generates a batch from four classes:

- *Hub round trips*, timed to stranded demand’s availability windows: a pickup toward the nearest hub, a delivery from it, proposed as a coordinated pair when neither endpoint is itself a hub.
- *Direct point-to-point rotations*, sampled without replacement with probability proportional to proximity and to the pair’s share of its origin’s unserved volume — close, locally dominant flows are the ones a direct can win.
- *Inter-hub trunk shuttles*, aimed at aggregate cross-hub tonnage after assigning every shipment’s endpoints to their nearest hubs; these feed all multi-hub itineraries at once and are therefore exempt from zone filtering.
- *Premium external capacity* as a last resort, for pairs no own-fleet proposal reaches; the model books it only if the demand still pays at that price.

Batch quotas keep lower-priority classes from being starved on dense instances, and deterministic keys (route plus departure bucket) deduplicate against everything alive or recently evicted. The extended instance is then re-solved with a warm-started column pool and the previous schedule seeded as incumbent — so a round can never return less than its predecessor — candidates are scored by whether they actually flew, and persistent non-flyers are evicted, which keeps the *active* model bounded while the *explored* candidate space grows without limit. Amnesty lets an evicted candidate return some rounds later, since the network context changes as accepted flights accumulate.

Two scale devices are toggleable: consolidation of tiny, far shipments into hub-corridor pseudo demands during the rounds — the final solve always runs on the original demand, so delivered schedules and profits stay exact — and zone rotation of proposal targeting, so each hub cluster receives a full batch and local synergies land in the same round. Algorithm 1 states the loop; full parameterization is in the repository’s technical report. One further class is a negative result we keep in the codebase, disabled: *hub waves*, coordinated feeder-trunk-distribution bundles proposed as a set for the long tail of scattered small shipments. The optimizer never adopted one (Section 8.5).

## 6 Geographic fix-and-optimize with exact gateway windows

Motivated by the observation that block-sized models adopt improvements in seconds while the whole-network master adopts nothing (Section 8.4), we decompose the planet: hub clusters (hubs within 3,000 km) define regions; every airport joins its nearest hub’s region.

**Blocks.** A block re-optimizes one region against a frozen world. Re-decidable (*kept*) flights are those with every leg inside the region, minus three repair classes computed to a fixpoint: (i) *balance repair* — the kept *selected* strings must balance arrivals and departures per (fleet, airport), or neither side of the partition assembles into rotations; unbalanced one-way strings

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**Algorithm 1:** Autonomous design loop (Section 5).

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input : base instance  $I_0$ ; batch size  $B$ ; round cap  $\bar{r}$ ; per-round budget; eviction window  $\bar{e}$ ; flat
        threshold  $\theta$ ; flat-round limit  $\bar{f}$  (with zones, at least one full zone cycle)
output: best schedule found, and the instance it flies on
 $I \leftarrow I_0$ ;  $\sigma \leftarrow \text{GREEDYCOVER}(I)$ ; // feasible, flowless seed
 $(\sigma, \Pi) \leftarrow \text{SOLVE}(I, \sigma, \emptyset)$ ; //  $\Pi$ : generated column pool
 $(I^*, \sigma^*) \leftarrow (I, \sigma)$ ; flat  $\leftarrow 0$ ; empty  $\leftarrow 0$ ;
for  $r = 1, \dots, \bar{r}$  while flat  $< \bar{f}$  do
     $U \leftarrow$  unserved tonnage of  $\sigma$  per shipment, ranked by revenue at risk;
     $C \leftarrow \text{PROPOSE}(I, U, B, \text{zone}(r), \text{blocked keys})$ ; // hub / direct / trunk / external
    if  $C = \emptyset$  then stop if ++empty completes a full zone cycle, else continue;
    empty  $\leftarrow 0$ ;  $(\sigma^+, \Pi) \leftarrow \text{SOLVE}(I \cup C, \sigma, \Pi)$ ; // seeded incumbent, warm columns
    if  $\sigma^+ = \perp$  then drop  $C$ , prune  $\Pi$  to  $I$ 's ids, continue;
    mark each  $c \in C$  flown in  $\sigma^+$ ; evict candidates unflown for  $\bar{e}$  consecutive rounds;
     $I \leftarrow (I \cup C) \setminus \text{evicted}$ ; remap  $\Pi$  and  $\sigma^+$  to the new ids;
    log (bound uplift of  $C$ , profit adopted); // promise vs. adoption
    if profit( $\sigma^+$ ) improves profit( $\sigma^*$ ) by less than  $\theta$  then flat  $\leftarrow$  flat + 1 else flat  $\leftarrow 0$ ;
    if profit( $\sigma^+$ ) > profit( $\sigma^*$ ) then  $(I^*, \sigma^*) \leftarrow (I, \sigma^+)$ ; // snapshot the instance  $\sigma^*$ 
    flies on
    ;
     $\sigma \leftarrow \sigma^+$ ;
/* final solve: last instance plus rescued ever-flown evicted candidates, full
   unconsolidated demand, seeded with the last schedule */
 $\sigma_f \leftarrow \text{EXACTFINAL}(I \cup \{\text{evicted } c \text{ flown at least once}\}, \sigma)$ ; // one fresh retry on
failure
return  $\sigma_f$  and its instance if the final solve succeeds (an exact final replaces the coarse best),
else  $(I^*, \sigma^*)$ 

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(typically a multi-stop trunk whose mirror has an en-route stop in a third region) are frozen one at a time, so the cascade follows the unbalanced chain and stops; (ii) flows whose path traverses the *kept set* on more than one maximal run, and (iii) degenerate cuts (a run entering and leaving the region at the same airport), whose flights are frozen because the merge splices exactly one regional segment per donated flow. Class (ii) is deliberately stated on the kept set rather than on geography, and recomputed inside the same fixpoint as balance repair: repair removes intra-region flights from the kept set while they keep operating, which can split a geographically single crossing into two kept runs — exactly the case the donation step must not see.

**Exact windows.** Cross-boundary cargo is truncated at its gateway with windows read from the *frozen timetable*, not estimated: an outbound segment must end before its frozen onward connection departs minus the gateway transfer time; an inbound segment becomes available when its frozen feeder lands plus transfer; a run starting at the shipment's true origin is released at availability plus loading, and one ending at the true destination is due at the deadline minus unloading. Cargo handling at true origins and destinations is thereby encoded into the windows so gateway hand-offs do not pay it twice. Consequently a time-feasible block solution *splices* back into a time-feasible global schedule by construction. Shipments with both endpoints inside the region enter whole; under deliver-all their contracted remainder rides along, so every block is seeded with its share of the incumbent — feasible for the block whenever the learned fleet correction (below) has not shaved the slice beneath the seed's own aircraft count.

**Fleet slice.** The block's fleet budget is  $\text{count} - \text{used}(\text{all}) + \text{used}(\text{kept})$ , computed by assembling the kept side alone (possible after balance repair). Splitting mixed rotation chains can cost a

rounding aircraft per fragment that no a-priori budget foresees; instead of estimating it, the cycle *learns* it: a merged block that fails the global fleet-size check by  $e_k$  aircraft teaches its block a budget reduction of  $\max(\cdot, e_k)$  for the next visit. Two honest caveats. The correction is deliberately *not* clamped to the seed’s own aircraft count: with a fully used fleet the idle slack is zero and any such clamp would permanently disable the learning — the shave exists precisely because the un-shaved budget produced infeasible merges. A slice below the seed’s count instead makes the block reject its own starting point and run unseeded; that costs a warm start, never feasibility (the merge guard). And the max update is a no-op when a repeat failure overshoots by less than the current correction; the cycle then simply keeps the incumbent for that block — the rule is a heuristic patch, not a convergence device.

**Monotone merge.** A block solution is spliced into the global schedule (frozen strings and flows kept; donated segments replaced; sub-contracted remainders fall back to global end-to-end contracting) and accepted *only if* the independently verified global profit improves. The cycle can therefore never lose ground (Proposition 2).

Algorithm 2 states block construction and the cycle. Two properties make the scheme safe; both are immediate, and we state them because they are what a reader must be able to check.

**Proposition 1** (Splice feasibility). *Let  $\sigma$  be a globally feasible schedule and let a block be built as in Algorithm 2. If  $\sigma_B$  is a feasible solution of the block instance, then every shipment of the merged schedule remains time-feasible: each donated segment is replaced by one whose elapsed time lies in  $[\alpha, \beta]$ , where  $\alpha$  is the arrival of the frozen feeder plus the gateway transfer time (the shipment’s availability plus loading when the run starts at its true origin) and  $\beta$  the departure of the frozen onward connection minus the transfer time (the deadline minus unloading when the run ends at the true destination), all read from  $\sigma$ ’s unchanged timetable.*

*Proof.* Frozen flights keep their departure and arrival times, so  $\alpha$  and  $\beta$  are unaffected by anything the block decides; at a true origin or destination the corresponding endpoint is the shipment’s own release or deadline, which no block decision can move either. A block shipment’s release and deadline are set to exactly  $\alpha$  and  $\beta$  by construction, and  $\sigma_B$  respects them; splicing therefore preserves every connection time on both sides of each gateway. It remains to classify every incumbent flow. Flows of shipments with both endpoints in the region are re-decided *in full*: the merge drops them entirely and installs the block’s routing, so they never ride a half-touched path (this covers, in particular, a whole-region shipment whose incumbent path detoured outside the kept set). Every other flow either donates its single kept run — replaced under the window argument above — or was frozen by the partition step, which freezes every flow with more than one maximal run through the kept set and every degenerate cut; because that criterion is evaluated on the *kept set* and re-applied within the same fixpoint as balance repair, at the fixpoint each remaining flow has at most one kept run and all its other legs are frozen.  $\square$   $\square$

One degenerate case is intentionally left to the runtime guard rather than the construction: a donated run whose exact window closes to a non-positive length is skipped (no sub-shipment is created) while its flights remain re-decidable; a block that then drops them produces a merge the independent check behind Proposition 2 rejects.

**Proposition 2** (Monotonicity). *Started from an incumbent that is feasible for the original instance, the regional cycle never decreases the global objective, and every schedule it returns is feasible for the original instance.*

*Proof.* Both statements are loop invariants. They hold at entry: the objective trivially, feasibility by hypothesis (the caller hands the cycle a feasibility-checked incumbent). Each iteration preserves them: a merged schedule is adopted only if it passes the independent feasibility

---

**Algorithm 2:** Geographic fix-and-optimize cycle (Section 6).

---

```

input : instance  $I$ , feasible incumbent  $\sigma$ , regions  $\mathcal{R}$ , per-block and total budgets
output: a schedule at least as good as  $\sigma$  (Proposition 2)
shave  $\leftarrow 0$ ; // learned fleet corrections, per block
foreach cycle, and pass  $R$  over  $\mathcal{R}$  then over cross-loaded region pairs do
    /* partition: which flights may this block re-decide? */
     $\mathcal{K} \leftarrow \{f \in F : \text{every leg of } f \text{ lies inside } R\}$ ;
    repeat
        foreach (fleet, airport) with unbalanced kept selected strings do
            | freeze one contributing one-way string;
        freeze the flights of flows with more than one maximal run through  $\mathcal{K}$ , and of
        degenerate cuts;
    until  $\mathcal{K}$  unchanged;
    if  $\mathcal{K}$  contains no cargo flight then continue;
    /* demand: exact windows read from the frozen timetable */
    foreach shipment  $q$  do
        if both endpoints of  $q$  lie in  $R$  then
            | add  $q$  whole, window shrunk by handling at both ends;
        else if  $\sigma$  routes  $q$  through  $\mathcal{K}$  on exactly one run  $[s, e]$  then
            | add a sub-shipment for that run, with release  $\alpha$  and deadline  $\beta$  as in Proposition 1;
     $n_k^B \leftarrow n_k - \text{used}_k(\sigma) + \text{used}_k(\mathcal{K}) - \text{shave}_k[R]$ ; // slice; may reject the seed (see
    text)
    seed  $\leftarrow$  kept strings and their flows; remainder contracted;
     $\sigma_B \leftarrow \text{SOLVE}(\text{block}, \text{seed})$ ;  $\sigma' \leftarrow \text{MERGE}(\sigma, \sigma_B)$ ;
    if  $\sigma'$  feasible and  $\text{profit}(\sigma') > \text{profit}(\sigma)$  then
        |  $\sigma \leftarrow \sigma'$ ;
    else
        | if  $\sigma'$  exceeds fleet  $k$  by  $e_k$  aircraft then  $\text{shave}_k[R] \leftarrow \max(\text{shave}_k[R], e_k)$ ;
return  $\sigma$ 

```

---

check *and* strictly improves the globally recomputed objective; otherwise the incumbent is kept unchanged. They therefore hold of the returned schedule in particular.  $\square$   $\square$

Proposition 1 is what the construction *intends*; Proposition 2 is what the implementation *guarantees* even when that intent is violated by a defect. Keeping both earned its keep: during development the guard rejected four distinct merge defects (Section 8.4) without ever admitting a corrupted schedule.

**Pairwise relay.** Optionally, after the single-region passes of each cycle, *pair* passes run on region unions ordered by open cross-tonnage. A cross-region shipment enters a pair block whole, with exact windows; feeder, trunk and distribution are decided in one model, avoiding any cross-pass ledger or estimated hand-off times.

## 7 Engineering for honest measurement

**Parallel pricing, bit-identical.** Each pricing iteration asks one independent shortest-path question per shipment against the same duals. The sweep runs one shipment per core and collects results in shipment order, so its output is bit-identical to the sequential sweep (asserted by test and verified in production runs: identical column counts and bounds). The per-destination A\* bounds — the only lazily built shared state — are prewarmed one Dijkstra pair per core. The servability probes of the design loop parallelize the same way with thread-local dual vectors.

| instance           | shipments | flights | nodes  | gap   | time (s) | termination    |
|--------------------|-----------|---------|--------|-------|----------|----------------|
| RC-I               | 800       | 100     | 3      | 0.00% | 0.1      | tree exhausted |
| IC-I               | 3,500     | 310     | 1      | 0.00% | 0.4      | tree exhausted |
| MI-I               | 4,000     | 400     | 13,698 | 1.66% | 180.0    | time limit     |
| EX-I               | 6,000     | 900     | 2,867  | 0.73% | 180.0    | time limit     |
| GI-I               | 9,000     | 1,250   | 1      | 0.43% | 97.6     | gap target     |
| RLA-I <sup>†</sup> | 29,819    | 1,250   | 1–3    | 9.9%  | 5,400    | no improvement |

Table 3: The base branch-and-price-and-cut across instance families (set-I instances), short budgets, no matheuristic layer. Gaps were measured with the pre-certification bound estimator (Limitation (viii)). <sup>†</sup>RLA is the deliver-all integrator instance of Table 2, run inside the design pipeline (Section 8.2); it differs from the others in more than size.

**Mispricing-safe dual smoothing.** Optionally, pricers see  $\alpha \cdot \text{center} + (1 - \alpha) \cdot \text{duals}$  [7, 8] ( $\alpha = 0.7$ ). Two safeguards keep exactness: a smoothed pass that finds no columns triggers a re-price with the true duals before any conclusion, and validity-critical bounds are computed from true-dual passes only. A test asserts that the stabilized root converges to the same LP value as the plain one. We report no performance numbers for smoothing: it is implemented, verified and off by default, and its evaluation is deliberately left to the multi-seed battery — we flag this to avoid claiming a benefit we have not measured.

**Bounds under truncation.** Hard per-round budgets are made mathematically legal by a Farley-type bound [9]: on interruption the tightest valid bound seen across full pricing passes is returned, each computed as the RMP value plus the most optimistic contribution of *missing* columns. A membership filter is essential in practice: pricers legitimately return columns already in the master (nonbasic at bounds, positive reported reduced cost); counting them produces absurd bounds. Only columns not yet in the master contribute.

**Cost visibility.** Every colgen iteration reports its breakdown (pricing / column loading / LP re-solve seconds). This 20-line addition settled in minutes a question we had been guessing at for days (Section 8.6).

## 8 Computational results

*Protocol and disclosure.* All numbers below are from single runs on one machine (Apple Silicon, 10 cores; CPLEX 22.2 as LP/MIP backend, HiGHS as fallback), on instance RLA-I-s1 unless stated. They are exploratory: no variance estimates, one seed. We report them because their margins are large and their patterns repeated across the campaign; the multi-seed battery over all instance families is the declared next step. All runs are reproducible from the public repository: instances and every run output are regenerated deterministically from their integer seeds by the documented commands (raw logs themselves are not versioned; see the Reproducibility section). One further disclosure: the campaign, including Table 3, was measured before the dual-bound certification fix of Limitation (viii), so its gaps are estimates until the certified re-run.

### 8.1 Where the branch-and-bound tree earns its keep

Before reporting what fails, we report where the base method works. Table 3 solves one instance of each family with the plain procedure (column generation, cuts, branching, restricted-master heuristic; no matheuristic layer) under a short budget.

| stage                                           | profit (M)    | increment |
|-------------------------------------------------|---------------|-----------|
| greedy cover seed (all demand contracted)       | 129.9         | —         |
| + seed flow loader (one warm LP over the pool)  | 139.6         | +9.7      |
| + local-branching ball $k=60$                   | 142.3         | +2.7      |
| + adoption re-monetization (one warm LP)        | 143.3 / 143.5 | +1.0      |
| unrestricted restricted-master MIP, same budget | 139.6         | +0.0      |

Table 4: Round-0 incumbent waterfall on RLA-I-s1 (consolidated model, deliver-all). The re-monetization column shows two independent runs. The last row is the ablation: without the ball, the same heuristic budget adopts nothing.

Three regimes are visible. On the small families the LP relaxation is integral or nearly so and the tree closes in milliseconds. In the middle (MI, EX) the relaxation is genuinely fractional and the tree does the work it was designed for — thousands of nodes, gaps under 2%. At GI scale (9,000 shipments, 1,250 flights) the root relaxation is already good enough that *one* node reaches a 0.43% gap in 98 seconds: the tree is not failing there, it is unnecessary. Only at RLA scale does the picture change qualitatively.

We stress what this table does *not* establish. RLA differs from GI in more than shipment count — it carries the deliver-all service commitment, gravity demand over 60% of station pairs, nine hubs instead of four, and weekly lane recurrence — so these measurements locate the breakdown between the two families without isolating its cause. A controlled scaling series over one family is the experiment that would, and it is not yet run.

## 8.2 The whole-network integer layer stalls

At RLA scale the regime of Table 3 changes. From a shared incumbent at 139.7–139.8 M (produced by cover seed + loader), the whole-network solver — warm-started column pool, seeded incumbent, escalating ball, 90-minute budget — produced *zero* incumbent improvements, twice (independent runs at 1,800s and 5,400s produced the same +0). Branch-and-bound explored 1–3 nodes. This is the phenomenon the rest of the paper responds to: at this scale the exact tree is vestigial and even ball-restricted sub-MIPs drown in the full master’s LP.

## 8.3 The adoption pipeline

Table 4 and Figure 2 show the round-0 waterfall on the consolidated design model (25,451 demands after tiny-far consolidation). Each stage is enabled on top of the previous.

The best full-demand schedule obtained by the design pipeline alone reaches 142.8 M at a 9.9% gap against the run’s dual-bound estimate (the certification caveat, Limitation (viii), applies); in it, 22,011 of 29,819 shipments (46.0 kt, a 46.6 M bill) are contracted and only 58 of 154 aircraft fly. That anatomy — a long tail of  $\sim 2$ -tonne shipments scattered over  $\sim 150$  stations, the amber fans of Figure 1 — motivates both the wave experiment (negative, below) and the decomposition (positive, next).

## 8.4 Geographic decomposition

A/B protocol: one shared base incumbent, then the same additional budget spent either (A) continuing the whole-network solve or (B) running the regional cycle. Table 5 summarizes four consecutive runs (two clean A/Bs plus two earlier runs whose arm B was invalidated by merge bugs — in *every* run, including the buggy ones, the global feasibility guard held: no corrupted schedule was ever accepted).

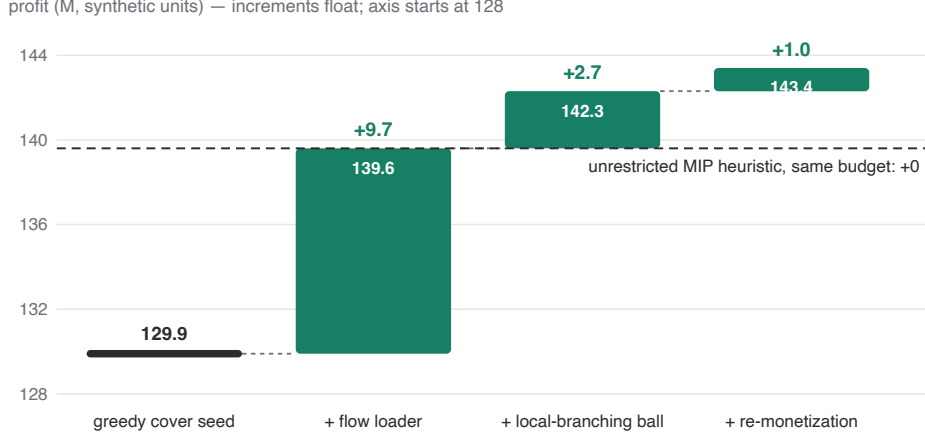


Figure 2: The adoption pipeline of Section 4 as a waterfall (data of Table 4). Green segments are the increments each stage adds on top of the previous; the dashed line is the ablation — the unrestricted restricted-master MIP, given the same budget as the ball, never leaves the loader’s level.

| arm                                     | budget  | gain    | note                         |
|-----------------------------------------|---------|---------|------------------------------|
| A: whole-network continuation           | 1,800 s | +0      |                              |
| A: whole-network continuation           | 5,400 s | +0      | independent repeat           |
| B: regional cycle                       | 900 s   | +5.03 M | 6 accepted blocks, then flat |
| B: regional cycle (learned fleet shave) | 1,800 s | +3.96 M | includes learning visits     |

Table 5: Whole-network vs. regional split from a shared incumbent (RLA-I-s1, full 29,819-shipment demand). Blocks of 200–350 flights adopt improvements in 4–50 seconds each.

The mechanism, not just the outcome, matches the diagnosis: all four regions adopt on their first visit (from +0.26 M to +1.5 M each, in 9–50s of block time), improvements shrink on the second cycle, and the cycle then honestly reports flat blocks until the budget closes. The best full-demand schedule of the campaign (144.8 M) came from a 5-minute base solve plus a 15-minute regional cycle.

## 8.5 Negative results

*Hub waves.* A fifth candidate class proposed coordinated feeder–trunk–distribution bundles per (hub corridor, day), with chain-consistent times and honestly windowed tonnage. Over 20 design rounds (6,000 candidates, waves included at 30% quota), *zero* were adopted, and the LP promise of candidate batches decayed from +4.4 M (round 1) to  $\sim 0$  (round 3 on). *Pair relay.* The pairwise passes of Section 6 ran clean (no crashes, all merges verified) and adopted nothing at  $\mu = 3$ . *Interpretation.* Three independent probes — waves, promise decay, pairs — agree: at triple own economics, the scattered cross-region tail is genuinely cheaper to contract than to fly. This is a statement about the instance’s economics, not about the algorithms; both mechanisms remain in the code, disabled by default, for cheaper-recourse or larger-scale settings.

*LP method.* On the colgen master ( $> 40k$  columns), CPLEX sifting — nominally suited to column-heavy LPs — measured  $2\text{--}3\times$  *slower* per re-solve (49–114s) than warm-started dual simplex (18–44s). The advantage of basis reuse across colgen iterations is well known; the value here is quantifying it for this master, since it rules out a drop-in remedy for the LP bottleneck of Section 8.6.

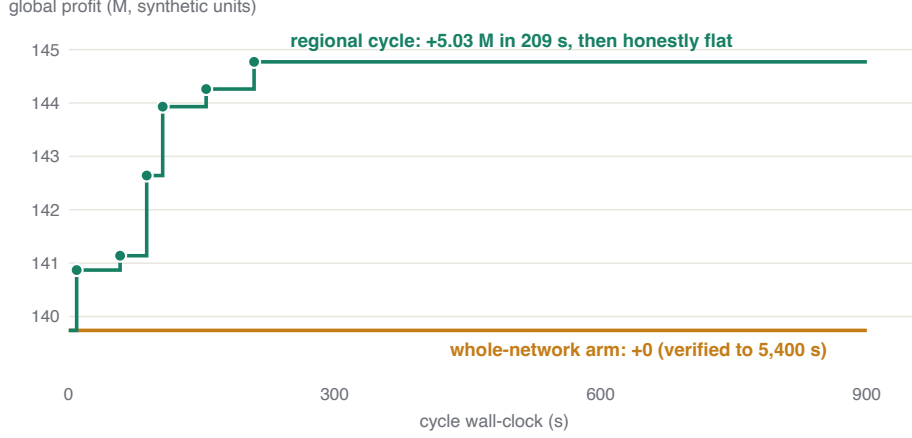


Figure 3: The clean A/B of Table 5, over the cycle clock. Each dot is an accepted block merge (globally verified); the plateau after 209 s is the cycle reporting flat blocks rather than manufacturing noise. The whole-network arm, from the same incumbent, adopts nothing in 5,400 s.

## 8.6 Parallel pricing and the next bottleneck

Parallelizing the pricing sweep cut it from  $\sim 25$  s to 0.4–1.0 s per iteration (identical columns and bounds), but end-to-end root time improved only  $\sim 15\%$  ( $156$  s  $\rightarrow$   $133$  s for six iterations). The per-iteration breakdown explains it: pricing 1.0 s, column loading 0.1 s, master LP re-solve 18–44 s and growing with the column count. Amdahl’s ledger is now unambiguous: the master LP is the next bottleneck, and the natural lever is column deletion (keeping the master lean) rather than a faster pricer. Dual smoothing also gains value from this shift: every avoided iteration now saves a full LP re-solve.

## 9 Limitations

(i) Headline results are single-seed, single-machine, one instance family at scale; margins are large but variances are unmeasured. (ii) All instances are synthetic; parameters (costs, demands, fleets, curfews, payload-range curves) are AI-estimated for plausibility and calibrated in scale — not in detail — to [1]; no real airline data is used or implied. (iii) Comparability with [1] is qualitative only: their instances and code were not published, so our reimplementations of their generators matches parameters, not files. (iv) There is no external baseline yet (e.g., a direct arc-flow MIP under equal budget); “better than our own previous configuration” is the only comparison this preprint supports. (v) The design loop is greedy across rounds; inner solves carry valid bounds, but no global optimality claim spans rounds. (vi) The breakdown of Table 3 is located between two instance families that differ in several factors at once; a controlled scaling series within one family is needed to attribute it to size. (vii) We do not compare against the concurrent model of [17]: the horizons differ (single night vs. periodic week) and neither instance set is expressible in the other’s formulation without changing the problem, so a head-to-head would compare problems rather than methods. (viii) A post-campaign code audit (September 2026) found and fixed two defects that affect the reported numbers, not the mechanisms. First, design-round instance rebuilds silently dropped the cargo-handling time (30 to 0 minutes), relaxing connection windows for the designed networks relative to the correctly handled baseline; the design-uplift margins are therefore optimistic until re-measured. Second, the dual bound combined a label-capped path pricer, which can miss an od’s best path, with a fleet-size string aggregation that strings crossing no fleet-count time escape, so pre-fix gaps

— including the 9.9% above and every gap of Table 3 — are estimates, not certificates. The solver now recomputes the Farley bound from an uncapped confirmation pass with a flight-cover string aggregation and stamps every run with a certification flag. A second audit pass (mid-September 2026) hardened the same account further: an exhausted search tree no longer reports the incumbent as a zero-gap certified bound when subtrees were pruned within the gap target (the honest certificate is  $\text{optimum} \leq \text{incumbent} \cdot (1 + \text{gap target})$ ); the maintenance string pricer’s week window was completed (a period-crossing predecessor could hide a feasible connection two week-slots back, and an unconstrained elapsed-time limit silently disabled string pricing altogether); and the rotation feasibility checker now accepts the week-wrapping inter-string connections the periodic master legally selects and prices. The multi-seed battery of the road map will run entirely on the fixed solver.

## 10 Conclusions and road map

At integrator scale, the exact tree of B&P&C contributes almost nothing, but the column-generation engine remains the value chain’s spine: every profitable move in our campaign flowed through priced columns, warm LPs over them, or dual signals derived from them. The productive question is not “exact or heuristic” but *where each layer earns its keep*: LP/colgen for columns, prices and certificates; local branching for integer moves the full master cannot digest; geographic fix-and-optimize for turning block-sized tractability into global, verified, monotone progress.

Road map, in order: (1) the multi-seed battery over all instance families with component ablations; (2) an arc-flow MIP baseline under equal budget; (3) column deletion for the master LP; (4) deeper trees inside regional blocks, where nodes cost seconds; (5) any form of real-data validation, which would upgrade every claim in this paper.

## Reproducibility and AI disclosure

The complete implementation (solver, generators, web workbench, 124 verification tests) and all instance generators are public in the accompanying repository; instances and every run output are regenerated deterministically from their integer seeds by the commands documented there — raw logs themselves are not versioned. This project was developed with substantial assistance from an AI coding agent (Anthropic Claude), including code, instance parameter estimation, experiment execution and the drafting of this manuscript; all algorithmic decisions, measurements and claims were verified by the reproducible pipeline above, and the human author reviewed and takes responsibility for the content. Airline names and data are fictional.

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